be summarized as follows:

1. The curriculum is designed in such a way so that the students can get, besides the regular theory and laboratory classes, the opportunity to develop software through separate software lab projects individually or in a group.

2. The curriculum includes two capstone courses, namely Design Project 1 and Design Project 2, where the students are involved in the development of a real life innovative project by integrating the knowledge of multiple areas.

3. The students need to take four selected elective courses in their senior years. Currently the selected elective courses are designed for four specialized areas of Software Engineering, namely network and system, software security, web development, and game development.

4. The program includes a 9.0 credit extensive Internship module where the students will be sent to software industries for about five months to equip them with the real life industry experience.

2.1 Program Educational Objectives (PEOs)

Graduates of the Computer Science and Engineering program are expected to attain the following objectives within a few years of graduation:

1. Demonstrate the ability to apply software engineering theories, models and techniques to analyze, design and develop the solution of real life problems.

2. Demonstrate professionalism, understand and carry the ethical values for the welfare of society, Muslim Ummah, and beyond.

3. Demonstrate strong awareness for life-long learning through self-motivation, professional training, and higher education.

4. Demonstrate the skill for effective communication, ability to interact with people of diverse educational and cultural background and work individually or in a team.

2.2 Program Outcomes (POs)

Students graduating from the Bachelor of Science in Software Engineering (B. Sc. in SWE) program, upon graduation, will have the ability to:

1. Engineering Knowledge: Apply the knowledge of mathematics, science, engineering fundamentals, and an engineering specialization to the solution of complex engineering problems.

2. Problem Analysis: Identify, formulate, research, and analyze complex engineering problems and reach substantiated conclusions using the principles of mathematics, the natural sciences, and the engineering sciences.

3. Design/Development of Solutions: Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for public health and safety and of cultural, societal, and environmental concerns.

4. Investigation: Conduct investigations of complex problems, considering experimental design, data analysis, and interpretation, and information synthesis to provide valid conclusions.

5. Modern Tool Usage: Create, select and apply appropriate techniques, resources, and modern engineering and IT tools, including prediction and modeling, to complex engineering activities with an understanding of their limitations.

6. The Engineer and Society: Apply reasoning informed by contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to professional engineering practice.

7. Environment and Sustainability: Understand the impact of professional engineering solutions in societal and environmental contexts and demonstrate the knowledge of and need for sustainable development.

8. Ethics: Apply ethical principles and commit to the professional ethics, responsibilities and norms of the engineering practice.

9. Individual Work and Teamwork: Function effectively as an individual and as a member or leader of diverse teams and in multidisciplinary settings.

10. Communication: Communicate effectively about complex engineering activities with the engineering community and with society at large. Be able to comprehend and write effective reports, design documentation, make effective presentations and give and receive clear instructions.

11. Project Management and Finance: Demonstrate knowledge and understanding of engineering and management principles and apply these to ones work as a team member or a leader to manage projects in multidisciplinary environments.

12. Life-Long Learning: Recognize the need for and have the preparation and ability to engage in independent, life-long learning in the broadest context of technological change.